

Assessment Schedule – 2024**Mathematics and Statistics: Demonstrate mathematical reasoning (91947)****Evidence**

Q	Evidence	Achievement	Merit	Excellence
ONE (a)	$2(3x - 5) + 2(2x + 3) = 56$ $6x - 10 + 4x + 6 = 56$ $10x - 4 = 56$ $10x = 60$ $x = 6 \text{ cm}$ OR alternative method: $3x - 5 + 2x + 3 = 28$ $5x - 2 = 28$ $5x = 30$ $x = 6 \text{ cm}$ Units not needed.	Correct answer, showing evidence of algebraic working.		
(b)	If $AB = y$, then $AC = y + k$; $BC = y + k - 4$ Perimeter $= 5y$ $y + y + k + y + k - 4 = 5y$ (1) $3y + 2k - 4 = 5y$ $2k - 4 = 2y$ $k - 2 = y$ Then perimeter $= 5y$ So perimeter $= 5(k - 2)$ Or perimeter $= 5k - 10$	Obtaining an equation involving y and k, i.e. reaching stage (1). OR Consistent expression for the perimeter, in terms of k. OR CAO.	Correct answer.	
(c)	Using triangle PBC: $PC^2 = PB^2 + BC^2$ $PC^2 = 12^2 + 10^2$ $PC^2 = 244$ $PC = \sqrt{244}$ $PC = 15.62 \text{ m}$ Using triangle TPC: $TC^2 = TP^2 + PC^2$ $TC^2 = 8^2 + 15.62^2$ $TC^2 = 308$ $TC = \sqrt{308}$ $TC = 17.55 \text{ m}$ OR alternative method: $TC = \sqrt{8^2 + 10^2 + 12^2}$ $= \sqrt{308} = 17.55 \text{ m}$	Finding the length PC. OR Finding length TB = 14.42. OR CAO.	Correct answer.	

(d)	Volume of one ball = 150 $\frac{4}{3}\pi r^3 = 150$ $\pi r^3 = 112.5$ $r^3 = 35.81$ $r = 3.296 \text{ and diameter} = 6.592$ Rectangular Box: Width = $2 \times 6.592 = 13.184 \text{ cm}$ Length = $3 \times 6.592 = 19.777 \text{ cm}$ Volume of the rectangular box $= 13.184 \times 19.777 \times 6.592 = 1718.87 \text{ cm}^3$ Empty space = $1718.87 - 6 \times 150$ $= 818.87 \text{ cm}^3$ Percentage empty space for the rectangular box $\text{box} = \frac{818.87}{1718.87} \times 100\% = 47.64\%$ Cylindrical Box: Radius = 3.296 cm Height = $12 \times 3.296 = 39.553 \text{ cm}$ Volume of the cylindrical box $= \pi \times 3.296^2 \times 39.553 = 1350 \text{ cm}^3$ Empty space = $1350 - 6 \times 150$ $= 450 \text{ cm}^3$ Percentage empty space for the cylindrical box $\text{box} = \frac{450}{1350} \times 100\% = 33.33\%$ i.e. cylindrical box has less empty space, with only 33.3% empty, compared to 47.64% empty in the rectangular box. OR alternative method: Total volume of rectangular box $= 6r \times 4r \times 2r$ Percentage filled space for rectangular box $= \frac{6(\frac{4}{3}\pi r^3)}{48r^3} = \frac{8\pi r^3}{48r^3} = \frac{8\pi}{48} = 0.5236 = 52.36\%$ Percentage empty space = 47.64% Total volume of cylindrical box $= \pi r^2 \times 12r = 12\pi r^3$ Percentage filled space for cylindrical box $= \frac{6(\frac{4}{3}\pi r^3)}{12\pi r^3} = \frac{8\pi r^3}{12\pi r^3} = \frac{8}{12} = 0.6666 = 66.66\%$ Percentage empty space = 33.33% i.e. cylindrical box has less empty space, with only 33.3% empty, compared to 47.64% empty in the rectangular box.	Calculating the radius of a single tennis ball. OR Consistent volume of rectangular box. OR Consistent volume of cylindrical box. OR Total volume of both boxes in terms of r.	- Empty space in the rectangular box. OR - Empty space in cylindrical box.	E7 / t1 - Empty space in the rectangular box. AND Cylindrical box AND Correct decision made, but no reference to percentages. OR Correct decision with minor error. E8 / t2 - Percentage empty space in a rectangular box. AND Percentage empty space in a cylindrical box. AND Decision made that cylinder has less empty space.
-----	---	---	---	---

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE question attempted towards solution.	1u	2u	3u	1r	2r	t1	t2

Q	Evidence	Achievement	Merit	Excellence
TWO (a)	$\sin \frac{x}{2} = \frac{5}{15}$ $\frac{x}{2} = \sin^{-1} \frac{5}{15}$ $\frac{x}{2} = 19.47^\circ$ $x = 38.94^\circ$	- Recognition that $\sin \frac{x}{2} = \frac{5}{15}$ - OR equivalent.	- Clear and justified working to show that $x = 38.94^\circ$. - Accept $x = 43.27^\circ$ if calculator is in grad	
(b)	In triangle CED: $\tan 33 = \frac{CD}{28}$ $CD = 28 \times \tan 33$ $CD = 18.18 \text{ m}$ $\text{Area of triangle CED} = \frac{1}{2} \times 28 \times 18.18$ $= 254.57 \text{ m}^2$ $\text{Area of triangle ABF also} = 254.57 \text{ m}^2$ $\text{Area of rectangle ACDF} = 30 \times 18.18 = 545.50 \text{ m}^2$ $\text{Area of stripe} = 545.50 - 2 \times 254.57 = 36.36 \text{ m}^2$ OR Alternative Method: In triangle CED: $\cos 33 = \frac{28}{CE}$ $CE = \frac{28}{\cos 33}$ $CE = 33.39 \text{ m}$ Let h be the distance between the parallel lines FB and EF: $\sin 33 = \frac{h}{2}$ $h = 2 \times \sin 33$ $h = 1.089 \text{ m}$ $\text{Area of stripe} = \text{area of parallelogram}$ $= 33.39 \times 1.089 = 36.36 \text{ m}^2$	- Finding length CD, with evidence of trigonometry. - OR - Consistent answer, with one minor error. - OR - Finding length CE, with evidence of trigonometry. - OR - CAO.	Correct value for the area of the stripe.	
(c)(i)	$\tan 60 = \frac{PF}{PB}$ $\tan 60 = \frac{PF}{k}$ $PF = k \times \tan 60$ $PF = 1.7321k$ OR Alternative Method: Because $\angle ABF = 60^\circ$ $\angle PBF = 30^\circ$ So $FB = 2k$ $PF = \sqrt{FB^2 + BP^2} = \sqrt{(2k)^2 + k^2} = 1.7321k$	Finding correct value for PF, with evidence of working.		

(ii)	Area ABFE = $\frac{1}{2} \times 1.7321k \times (2k + 4k)$ = $5.1963k^2$ Volume = 649.519 $5.1963k^2 \times 8k = 649.519$ $41.5704k^3 = 649.519$ $k^3 = \frac{649.519}{41.5704}$ $k^3 = 15.625$ $k = \sqrt[3]{15.625}$ $k = 2.5 \text{ m}$ Units not required. Accept any appropriate rounding.	• Correct expression for area of ABFE. OR Sensible steps of working towards finding k, from an incorrect expression for ABFE. OR CAO.	• Consistent value for k, following from error in area of ABFE. OR Correct value of k, but without evidence of clear working.	E7 / t1 • Correct value of k, with a minor error, but with a clear working. E8 / t2 • Correct value of k, with evidence of clear working.
------	---	--	---	---

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE question attempted towards solution.	1u	2u	3u	1r	2r	t1	t2

Q	Evidence	Achievement	Merit	Excellence
THREE (a)	$x = 0$ gives $y = 5^{0+2} + 4 = 29$ i.e. required point is (0,29) OR $y=29$	- Correct answer. - OR - CAO		
(b)(i)	$y = (x + 4)^2 - 1$ OR $y = x^2 + 8x + 15$ OR $y = (x + 3)(x + 5)$	CAO	- Correct equation, with some justification. - Accept n instead of x. - Students do not need $y =$	
(ii)	Features are : Crosses x -axis at $(-3,0)$ and $(-5,0)$. Crosses y -axis at $(0,15)$. Minimum at $(-4,-1)$. Line of symmetry of $x = -4$. Graph will be a positive parabola. Range of $y \geq -1$. Accept intercepts as $y=15$ and $x=-5, -3$	One non-trivial feature.	At least three non-trivial features.	
(c)(i)	$y = -1.5x + 11.5$ Or equivalent.	Correct equation.		
(ii)	Algebraic Method : At the intersection points: Solve simultaneously: $y = -1.5x + 11.5$ and $x - 2y - 9 = 0$ Giving $x - 2(-1.5x + 11.5) - 9 = 0$ $x + 3x - 23 - 9 = 0$ $4x = 32$ $x = 8$ Then $y = -1.5 \times 8 + 11.5$ $y = -0.5$ i.e. intersection at (8,-0.5) Allow consistency from (c) (i) Allow alternative valid algebraic methods. OR Graphical Method : Draws accurate graph of $x - 2y - 9 = 0$ Intersection at $(8,-0.5)$ identified. Evidence must be shown that the co-ordinates have been found from a graphical method and their graph.	- Line G not drawn accurately and then with a consistent intersection solution. - OR - Draws accurate graph of Line G $x - 2y - 9 = 0$. - OR - Algebraic simultaneous equations, with elimination of one of the variables. - OR - Solution found from a guess and check method. - OR - CAO.	- Intersection point identified from clear algebraic evidence. - OR - Intersection point identified from evidence of a graphical method.	

(d)	$3^{2x+3} \times 3^{2x} = 3^{4-3y}$ # (1) $3^{2x+3+2x} = 3^{4-3y}$ $3^{4x+3} = 3^{4-3y}$ $4x + 3 = 4 - 3y$ # (2) $3y = 1 - 4x$ $y = \frac{1 - 4x}{3}$	• Correctly reaching stage #(1).	• Correctly reaching stage #(2).	E7 / t1 • Correct expression for x. OR Correct expression for y , with a minor error. E8 / t2 • Correct expression for y.
-----	--	---	---	---

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE question attempted towards solution.	1u	2u	3u	1r	2r	t1	t2

Cut Scores

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 6	7 – 13	14 – 20	21 – 24

Solution: Question Three (c)(ii)

